Principles of Effective Antibiotic Therapy

Lecture Handout — Slides 1–62

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2026-02-25

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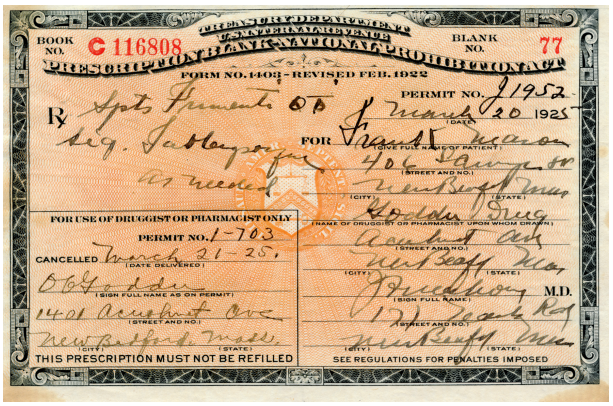
1 Introduction: Antibiotics — The Medical Miracle

1.1 Medicine in the Pre-Antibiotic Era

Prior to the availability of antibiotics, medicine was largely a process of **prognostication** rather than intervention. Physicians could predict the course of infectious diseases but had little ability to alter outcomes. Treatments were rudimentary and often ineffective — prescriptions frequently consisted of alcohol-based preparations such as *spiritus frumenti* (whiskey) or high-proof herbal tonics like Jamaican ginger (90% alcohol).



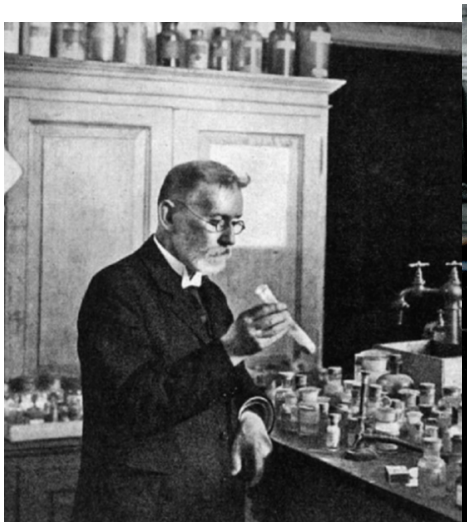
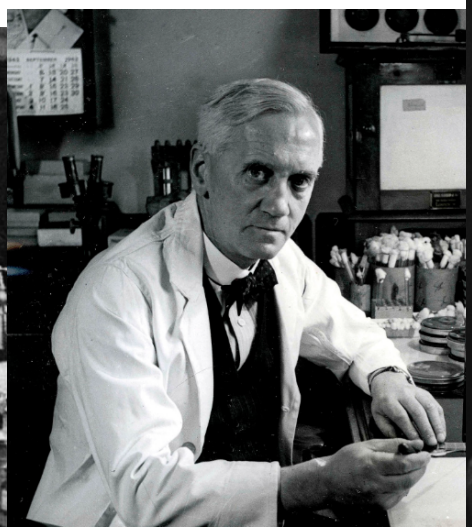
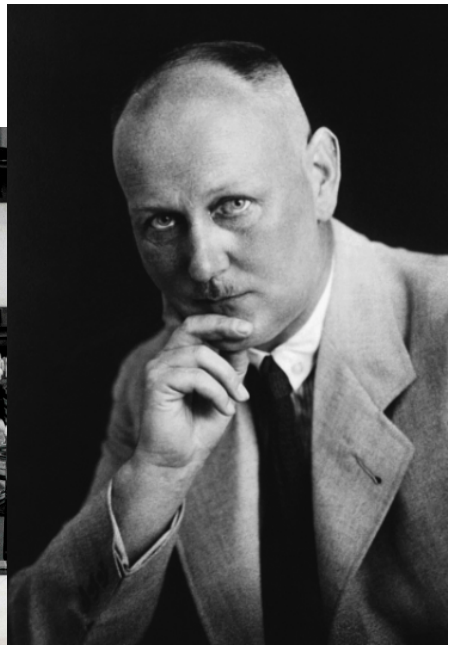
(a) A doctor's prescription from Massachusetts (1925) for *spiritus frumenti* — whiskey — written in Latin to obscure the contents from the patient.

(a) Jamaican ginger tonic (90% alcohol) marketed for dyspepsia, cholera morbus (gastroenteritis), colic, cramps, and flatulence.

1.2 Dawn of Antibiotic Discovery

Although evidence suggests antibiotic-like substances were used by ancient civilizations — Egyptians produced tetracycline-containing beer using *Streptomyces*-contaminated grain, and the Greeks and Chinese employed various antimicrobial remedies — the first truly effective systemic antibiotics emerged in the early 20th century.

Table 1: The three founding figures of modern antibiotic therapy.

Paul Ehrlich (1909) Salvarsan — first treatment for syphilis; highly toxic, required skilled administration	**Alexander Fleming** (1929) Penicillin discovery; purified and tested by Florey, Chain & Heatley (1940)	**Gerhard Domagk** (1931) Sulfonamides — discovered through textile dye research; treated his own daughter's severe cellulitis before human trials
		

1.3 Mortality Reduction with Antibiotic Therapy

The impact of antibiotics on mortality from common infectious diseases is unparalleled by any other class of therapeutic drugs — far exceeding the impact of treatments for myocardial infarction, stroke, or cancer.

Table 2: Comparison of mortality rates before and after the availability of antibiotic therapy. Adapted from Spellberg (Spellberg, 2025).

Disease	Pre-antibiotic era	Antibiotic era	Mortality reduction
Community-acquired pneumonia	~35%	~10%	**−25%**
Nosocomial pneumonia	~60%	~30%	**−30%**
Bacterial endocarditis	~100%	~25%	**−75%**
Gram-negative bacteremia	~70%	~10%	**−60%**
Bacterial meningitis	>80%	<20%	**−60%**
Cellulitis	~11%	<0.5%	**−10.5%**

1.4 Antibiotics: Essential for Modern Medicine

Beyond treating infections directly, antibiotics enable virtually every aspect of modern interventional medicine:

- **Complex surgery** — deeply invasive and prolonged surgical procedures
- **Cancer chemotherapy** — aggressive cytotoxic regimens that suppress immunity
- **Critical care** — maintenance of patients with central venous catheters, mechanical ventilation
- **Neonatal medicine** — care for premature infants and mothers with post-partum sepsis
- **Transplantation** — solid organ and stem cell transplantation

  Antibiotics created a revolution in the practice of medicine, transforming a primarily diagnostic-focused field into a *therapeutic, interventional* profession. (Spellberg, 2025)

Without effective antibiotics, medicine would regress approximately 100 years — losing the ability to safely perform these high risk procedures.

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2 The Antimicrobial Resistance Crisis

2.1 Global Impact of Resistance

Antimicrobial resistance (AMR) represents one of the most significant threats to global public health. Current global statistics reveal the scale of the problem:

- **21.6 million** deaths due to sepsis annually
- **7.75 million** deaths due to bacterial infections
- **4.7 million** deaths *associated* with resistance (requiring less effective or more toxic antibiotics)
- **>1 million** deaths directly *attributable* to resistance

In **Italy** specifically, approximately **8,000 deaths per year** are attributable to antibiotic-resistant infections — a figure that receives remarkably little public attention relative to its magnitude.

2.2 WHO Priority Pathogen List

The World Health Organization has classified resistant pathogens into three priority tiers to guide research and development of new antibiotics:

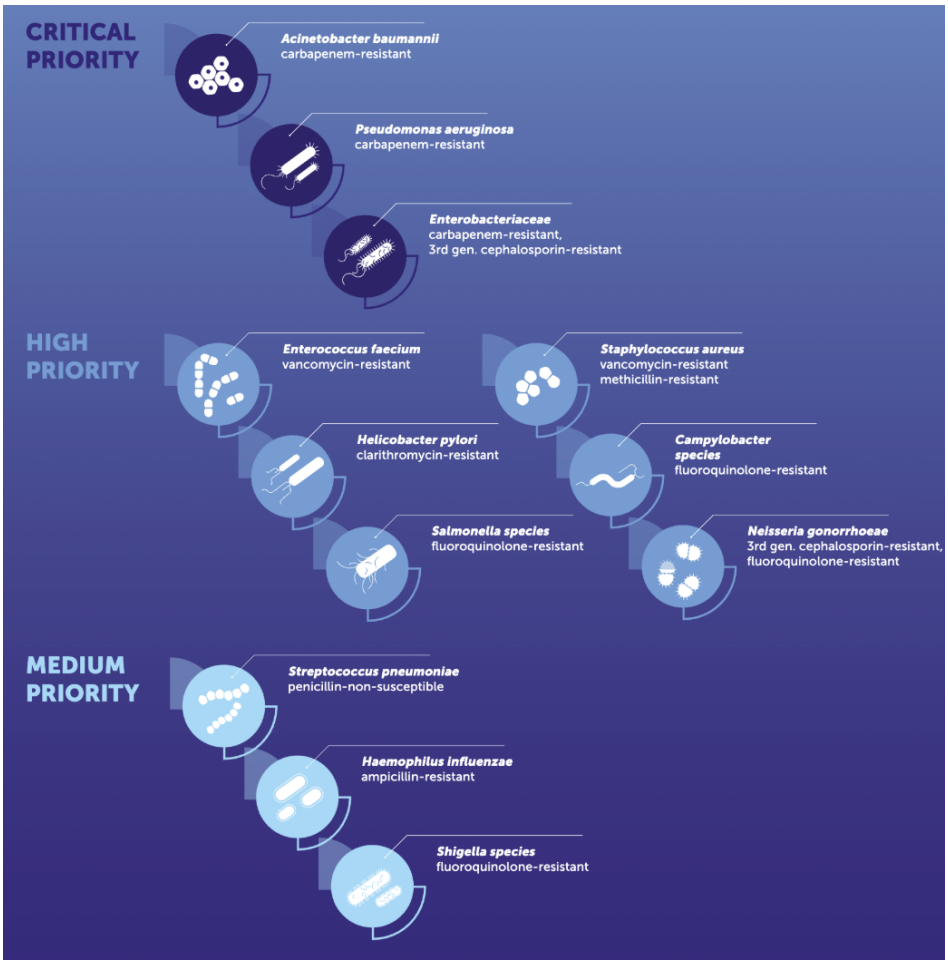

Figure 3: WHO Bacterial Priority Pathogens List, organized by critical, high, and medium priority.

2.2.1 Critical Priority (all Gram-negatives)

These are pathogens typically acquired by hospitalized, immunocompromised, or critically ill patients, often emerging after multiple courses of antibiotic therapy:

- ***Acinetobacter baumannii*** — often begins as colonization in ventilated or burn patients; progresses to pneumonia, bloodstream infection, and sepsis; frequently multi-drug resistant (MDR) and sometimes pan-drug resistant (PDR)
- ***Pseudomonas aeruginosa*** — associated with immunocompromised and critically ill patients; commonly MDR
- **Carbapenem-resistant Enterobacterales** — *E. coli*, *Klebsiella pneumoniae*, *Enterobacter* spp. — the most common causes of urinary tract infections and bloodstream infections, now increasingly resistant to carbapenems (previously considered last-line agents)

2.3 Resistance Rates in Italy and Europe

2.3.1 Methicillin-resistant *Staphylococcus aureus* (MRSA)

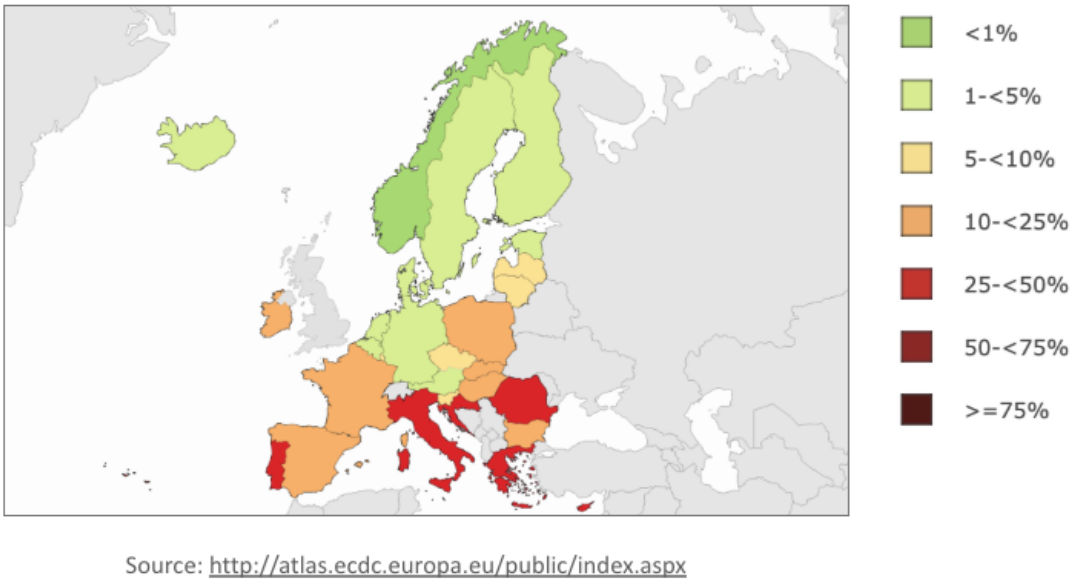

Figure 4: European MRSA rates — Italy falls in the 25–50% range, requiring empirical MRSA coverage for virtually all suspected staphylococcal infections.

In Italy, **25–50% of *S. aureus* isolates are methicillin-resistant**, necessitating the use of intravenous antibiotics such as vancomycin or daptomycin that are more toxic, more expensive, and require prolonged hospitalization compared to oral alternatives available for methicillin-susceptible strains.

2.3.2 Carbapenem-resistant Enterobacterales (CRE)

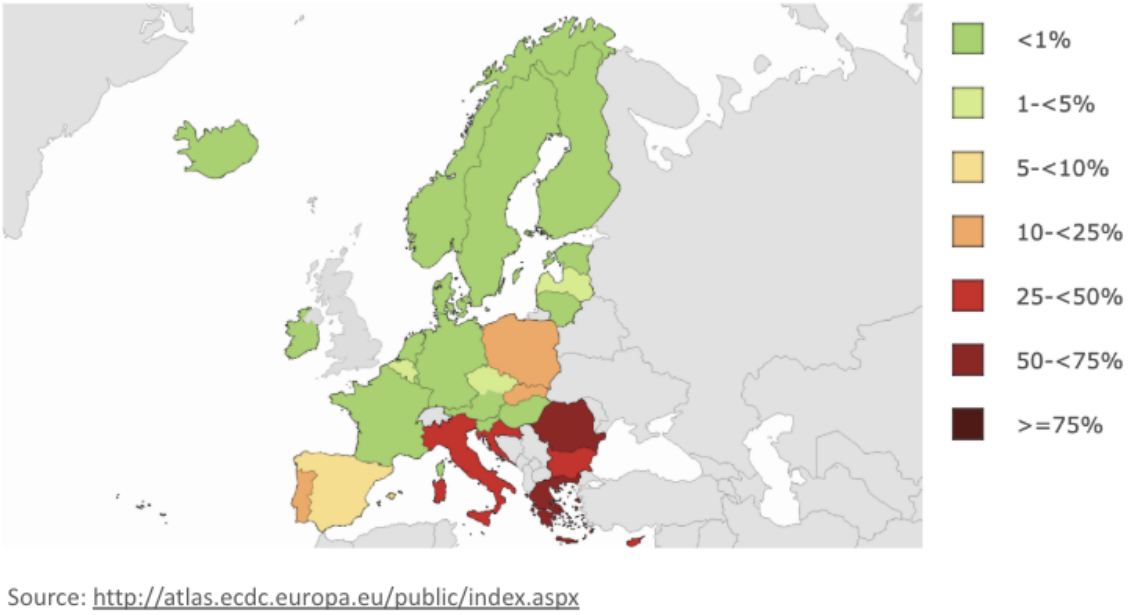

Figure 5: European CRE rates — Italy is in the 25–50% range; Greece and Turkey exceed 50%.

Up to 25% of *Klebsiella pneumoniae* hospital isolates in Italy are resistant to carbapenems. This has transformed the mortality of common Gram-negative bloodstream infections from approximately 8–10% to **40–50%**, because physicians must resort to less effective combination regimens.

2.3.3 Multi-drug Resistant *Acinetobacter baumannii*

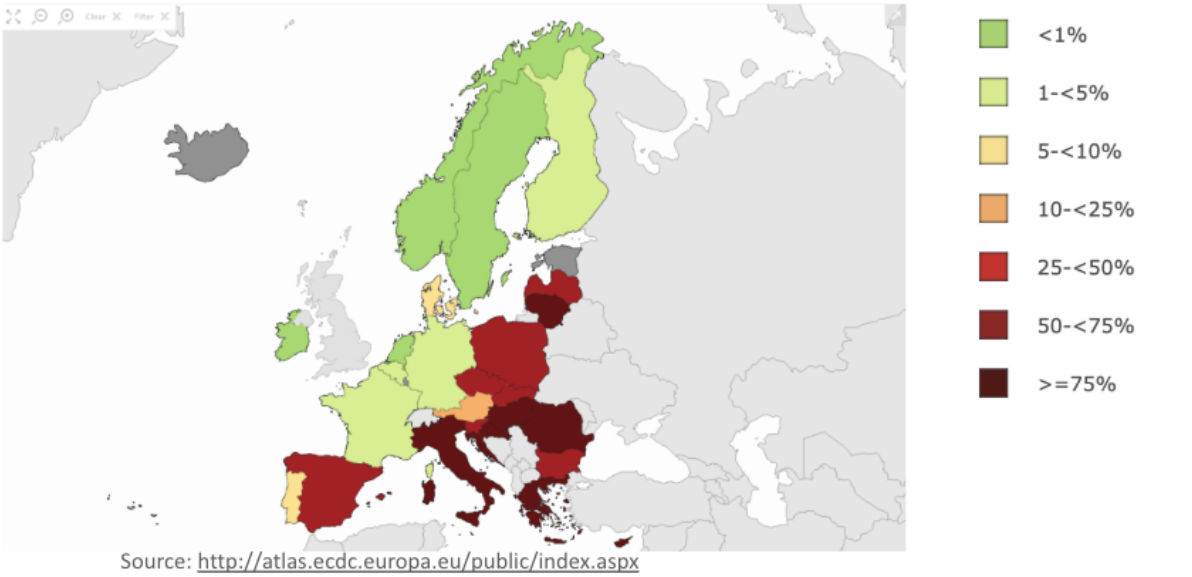

Figure 6: European MDR *A. baumannii* rates — Italy falls in the 50–75% range.

Why are Northern European countries doing better?

Northern European countries have lower resistance rates due to: (1) lower per-capita antibiotic use, (2) stricter infection control practices and culture, (3) modern hospital infrastructure with fewer multipatient rooms. However, resistance travels with people — international travel and migration disseminate resistant organisms globally.

For example: Antibiotic consumption (defined daily doses or DDD per 1,000 inhabitants per day):

- Netherlands, Sweden: ~8–12
- Germany, Denmark: ~10–14
- Italy, Spain: ~18–25
- Greece: ~30–35

2.4 The Antibiotic Discovery Pipeline Is Slowing

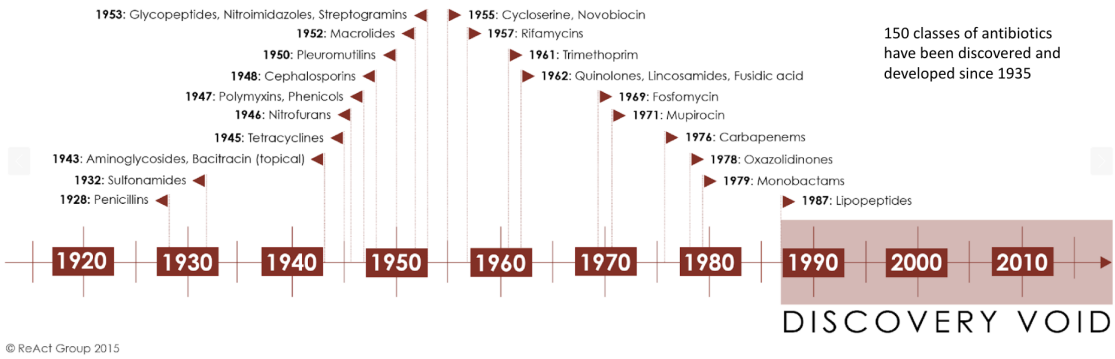

Figure 7: Timeline of antibiotic class discovery — a "discovery void" has persisted since the 1980s.

Antibiotic development is declining because:

- **High cost and risk** — developing a new antibiotic costs approximately $1 billion
- **Poor return on investment** — hospitals appropriately restrict new antibiotics to preserve them, reducing sales
- **Competition with profitable drug classes** — antibiotics cannot compete commercially with GLP-1 inhibitors, statins, or other drugs taken daily for life
- **No novel mechanisms** — recent approvals are iterations of existing classes, not truly new drug classes

New funding models are needed. Italy has passed measures supporting a "Netflix model" — paying a subscription fee to ensure antibiotics are available when needed, regardless of usage volume. Antibiotics can also be thought of like fire extinguishers: you hope you never need them, but someone must pay to keep them functional (see this blog by Dr. John Rex for more information.

2.5 Antibiotics Are a Societal Trust

> Antibiotic overprescription is a ***tragedy of the commons***: an individual undertakes an action perceived to be in their own self-interest that causes harm to society at large.

- **>1/3 of community antibiotic prescriptions** are inappropriate
- **Up to 50% of hospital antibiotic prescriptions** may be inappropriate
- Each inappropriate prescription individually causes minimal harm, but collectively the effect is catastrophic

One of the most important functions of the physician is to serve as an expert steward — both using antibiotics effectively and protecting their efficacy for future patients.

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3 The Ten Principles of Effective Antibiotic Therapy

! Overview of the 10 Principles

1. Develop an accurate differential diagnosis
2. Only use antibiotics when they alter the clinical course
3. Empirically target microbes in the differential diagnosis
4. Use a lower threshold for empirical therapy in critically ill patients
5. Host factors affect the spectrum of empirical therapy
6. Use PK/PD principles to optimize treatment selection and dosing
7. De-escalate therapy based on microbiology results and clinical response
8. If therapy is not working, consider source control or alternative diagnosis before broadening
9. Distinguish new infection from failure of initial therapy
10. Duration of therapy should be as short as possible based on evidence

This handout covers Principles 1–6 in detail, as presented in the lecture.

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4 Principle 1: Develop an Accurate Differential Diagnosis

4.1 The Medical History Is 80% of Diagnosis

In infectious diseases, the **patient's medical history** — not imaging, biomarkers, or even microbiological tests — is the single most important diagnostic tool. This may seem counterintuitive, but there are important reasons:

- Microbiological tests, while confirmatory, are **not perfectly sensitive or specific**
- Ordering tests in patients with low pre-test probability leads to **false positives** that can trigger unnecessary invasive procedures or toxic treatments
- Broad-panel testing without clinical context produces **incidental findings** that mislead clinicians

4.1.1 Key Elements of the Infectious Disease History

- **Current symptoms** — assessed using the **8 cardinal descriptors**: Timing, Location, Character, Aggravating factors, Alleviating factors, Associated symptoms, Severity, Setting
- **Fever** — height, duration, pattern (continuous, intermittent, relapsing)
- **Risk factors** — indwelling devices (urinary catheters, vascular catheters, prosthetic joints, cardiac devices), recent procedures, immunosuppression, diabetes, injection drug use, previous infections
- **Travel and exposure history** — countries, regions (urban vs. rural), dates, animal or insect exposure, contact with ill individuals, consumption of potentially contaminated food/water
- **Sexual history** and STI risk
- **Vaccination history**
- **Past medical, surgical, medication history, and allergies**
- **Recent antibiotic use**
- **Social history** — long-term care facility residence, occupation, hobbies, substance use, unusual exposures

4.2 Case Example: The Importance of History

4.2.1 Clinical Presentation

A **34-year-old farmer from Sicily** presents with worsening back pain after sitting for more than a couple of hours. MRI reveals evidence of inflammation and septic discitis at the L4 vertebral body, suggesting spondylitis.

4.2.2 The History Changes Everything

Pertinent history reveals:

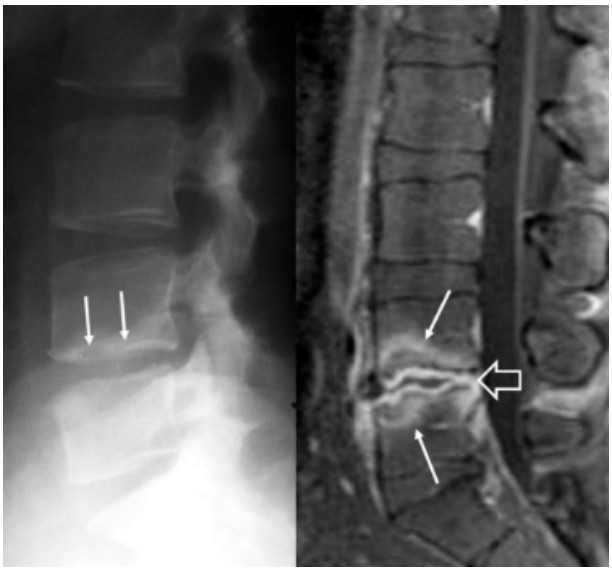
- Patient is a **sheep and cattle farmer** producing milk and cheese (Pecorino salato and Ricotta)
- Pain began after a **"bad case of flu"** with fever, achy joints, and headache

4.2.1.1 Standard Spondylitis Etiologies

The most common causes of vertebral osteomyelitis/spondylitis are:

- **>50%**: *Staphylococcus aureus*, *S. epidermidis*
- **~25%**: *Streptococcus* spp., *Enterococcus* spp., *Pseudomonas aeruginosa*, Enterobacterales, anaerobes, *Mycobacterium tuberculosis* (Pott's disease)

These are typically **hematogenously spread** infections, classically occurring in hospitalized patients after central venous catheter use or staphylococcal bacteremia.



(a) MRI showing septic discitis and bone marrow edema at L4–L5 vertebral bodies, consistent with spondylitis.

- **Father** (also works on the farm) has **worsening hip pain** — being evaluated for hip replacement
- From an **endemic area** for zoonotic infections in southern Italy

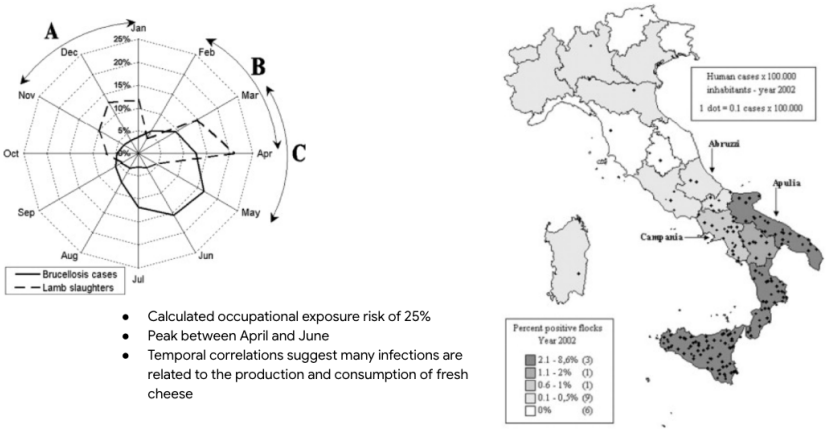


Figure 9: Brucellosis epidemiology in Italy — a zoonotic Gram-negative coccobacillus transmitted through contact with livestock and unpasteurized dairy products (Massis et al., 2005).

4.2.3 Diagnosis: Brucellosis Spondylitis

The standard diagnostic workup and treatment for staphylococcal spondylitis would **completely miss** this diagnosis:

Table 3: Comparison highlighting how history-driven diagnosis changes the entire diagnostic and therapeutic approach.

Aspect	Standard spondylitis	Brucellosis spondylitis
Blood cultures	Standard incubation	**Prolonged incubation** required; bone marrow biopsy may be needed

Aspect	Standard spondylitis	Brucellosis spondylitis
Specific tests	Not required	**Brucella PCR**, combination serologic studies
Treatment	Vancomycin or anti-staphylococcal agents	**Gentamicin + Doxycycline + Rifampin** — completely unconventional

Key teaching point: The history informed everything — from diagnostic test selection to treatment. Starting empirical vancomycin (the standard approach) would have been completely ineffective against Brucella.

4.3 Antibiotics Are Usually Started Empirically

In practice, antibiotic therapy is initiated **before** microbiological results are available, because the most critical window for treatment is the first 24–48 hours when bacterial inoculum is highest and patients are sickest.

4.3.1 Timeline from Suspicion to Definitive Therapy

Time	Step	Action
0 hours	Suspect infection	Clinical assessment
~1 hour	Culture suspected sites	**Begin empiric therapy** for likely pathogens
~24 hours	Gram stain results	Preliminary guidance for therapy adjustment
36–48 hours	Pathogen identification	Consider narrowing therapy
72–96 hours	Susceptibility results	**Switch to definitive therapy**

4.3.2 The Problem of Empirical Overuse

Because empirical therapy is started before diagnosis, this inherently leads to **overuse**:

- **>90% of upper respiratory tract infections** are viral and should not be treated with antibiotics
- Clinicians frequently **overestimate** the risk of bacterial infection
- Clinicians frequently **underestimate** the risks of antibiotic therapy

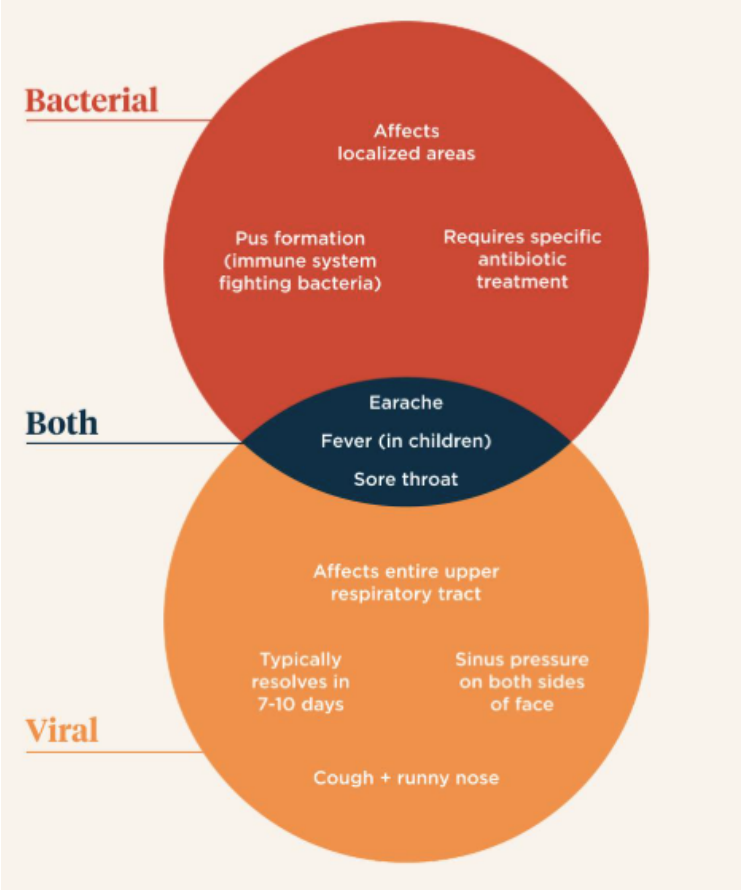


Figure 10: Overlap between bacterial and viral upper respiratory tract infections — the vast majority are viral and do not require antibiotics (Chow et al., 2012).

4.4 Risks of Antibiotic Therapy Are Underappreciated

Key Statistics

- **1 in 5 patients** given an antibiotic prescription develops an adverse event or superinfection (resistant pathogens or *Clostridioides difficile*)
- Each additional **10 days** of antibiotic therapy confers a **3% increased risk** of an adverse drug effect

This results from a combination of (1) **fear from diagnostic uncertainty**, and (2) **lack of appreciation for how dangerous antibiotics can be** (Tamma et al., 2017).

4.4.1 Most Common Antibiotic-Associated Adverse Events in Hospitalized Patients

Table 3. Proportion of 30-Day Antibiotic-Associated Adverse Drug Events in 1488 Hospitalized Patients Receiving Systemic Antibiotic Therapy[a]

Antibiotic Agent	No. of Patients Receiving Agent	No. (%)						
		Cardiac	Gastro-intestinal[b]	Hematologic	Hepato-biliary	Renal	Neurologic	Other Events[c]
β-Lactams[d]	1187	0	59 (5.0)	27 (2.3)	6 (0.5)	17 (1.4)	10 (0.8)	2 (0.2)
Ampicillin	63	0	2 (3.2)	1 (1.6)	1 (1.6)	1 (1.6)	0	0
Amoxicillin-clavulanate	102	0	3 (2.9)	0	0	0	0	0
Ampicillin-sulbactam	52	0	1 (1.9)	0	0	2 (3.8)	0	0
Oxacillin	33	0	4 (12.1)	1 (3.0)	2 (6.0)	0	0	0
Piperacillin-tazobactam	315	0	16 (5.1)	4 (1.3)	1 (0.3)	1 (0.3)	1 (0.3)	1 (0.3)
Cefazolin	79	0	0	1 (1.3)	0	2 (2.5)	0	0
Ceftriaxone	607	0	14 (2.3)	11 (1.8)	3 (0.5)	5 (0.8)	1 (0.2)	0
Cefpodoxime	89	0	2 (2.2)	0	0	0	0	0
Cefepime	414	0	10 (2.4)	6 (1.4)	0	6 (1.4)	7 (1.7)	1 (0.2)
Ertapenem	85	0	3 (3.5)	0	0	0	0	0
Meropenem	80	0	4 (5.0)	3 (3.8)	0	0	1 (1.3)	0
Non-β-lactams								
Aminoglycosides	32	0	0	0	0	2 (6.3)	0	0
Azithromycin	400	1 (0.3)	1 (0.3)	0	4 (1.0)	0	0	0
Clindamycin	193	0	3 (1.6)	0	0	0	0	0
Daptomycin	8	0	0	0	0	0	0	1 (12.5)
Doxycycline	57	0	2 (3.5)	0	0	0	0	0
Fluoroquinolones	394	1 (0.3)	5 (1.3)	1 (0.3)	3 (0.8)	1 (0.3)	1 (0.3)	1 (0.3)
Linezolid	23	0	0	0	0	0	1 (4.3)	0
Metronidazole	175	0	1 (0.6)	0	0	0	1 (0.6)	0
Trimethoprim-sulfamethoxazole	155	0	5 (3.2)	0	0	6 (3.9)	0	1 (0.6)
Intravenous vancomycin	544	0	2 (0.4)	0	0	19 (3.5)	0	2 (0.4)
Any antibiotics	1488[e]	2 (0.1)	78 (5.2)	28 (1.9)	13 (0.9)	45 (3.0)	13 (0.9)	7 (0.5)

[a] The following regimens are included in the overall rates and resulted in no 30-d adverse drug events: penicillin (21), amoxicillin (47), dicloxacillin (1), cephalexin (44), second-generation cephalosporins (38), ceftazidime (6), ceftaroline (8), aztreonam (22), fosfomycin (10), nitrofurantoin (26), tigecycline (3), oral vancomycin (84).

[b] Includes nausea, emesis, non–*Clostridium difficile*–associated diarrhea.

[c] Other adverse drug events include cefepime-associated anaphylaxis (1), piperacillin-tazobactam–associated drug fever (1), ciprofloxacin-associated tendinitis (1), daptomycin-associated myositis (1), trimethoprim-sulfamethoxazole–associated pancreatitis (1), vancomycin-associated hives (1), and trimethoprim-sulfamethoxazole-associated nonhives rash (1).

[d] Some patients received more than 1 β-lactam antibiotic.

[e] Most patients (1176 [79%]) received more than 1 antibiotic.

Figure 11: 30-day antibiotic-associated adverse events in hospitalized patients receiving systemic therapy. Gastrointestinal effects are most common, followed by hematological effects (thrombocytopenia, granulocytopenia in up to 2%), and renal or neurological toxicities. Certain drug classes (aminoglycosides, vancomycin) carry particularly high renal injury risk (Tamma et al., 2017).

4.5 Positive Cultures Are Not Always Proof of Infection

A positive culture in the **absence of signs and symptoms of infection** should **not** reflexively trigger antibiotic therapy. Without clinical symptoms, positive cultures often represent **colonization or contamination**.

Common offenders include:

Sample type	Problem
Superficial wound swabs	Sample skin flora, not the pathogen causing deep infection; deep tissue biopsy at infection margins is required
Urine cultures	Catheterized patients always have positive cultures; asymptomatic bacteriuria should not be treated
Bronchoalveolar lavage	Can become contaminated with oral flora during bronchoscope withdrawal
Respiratory samples	Low positive predictive value without compatible symptoms
Stool cultures	Unreliable without active diarrhea at time of collection

4.5.1 Diagnostic Stewardship Works

Simple interventions can dramatically reduce inappropriate testing. A study by Luu et al. placed reminders at nursing stations with key messages:

> *Stop sending urine cultures for everything nosocomial. Cystitis doesn't cause fevers. Urine from the catheter is not sterile. A positive urine culture does not equal infection.*

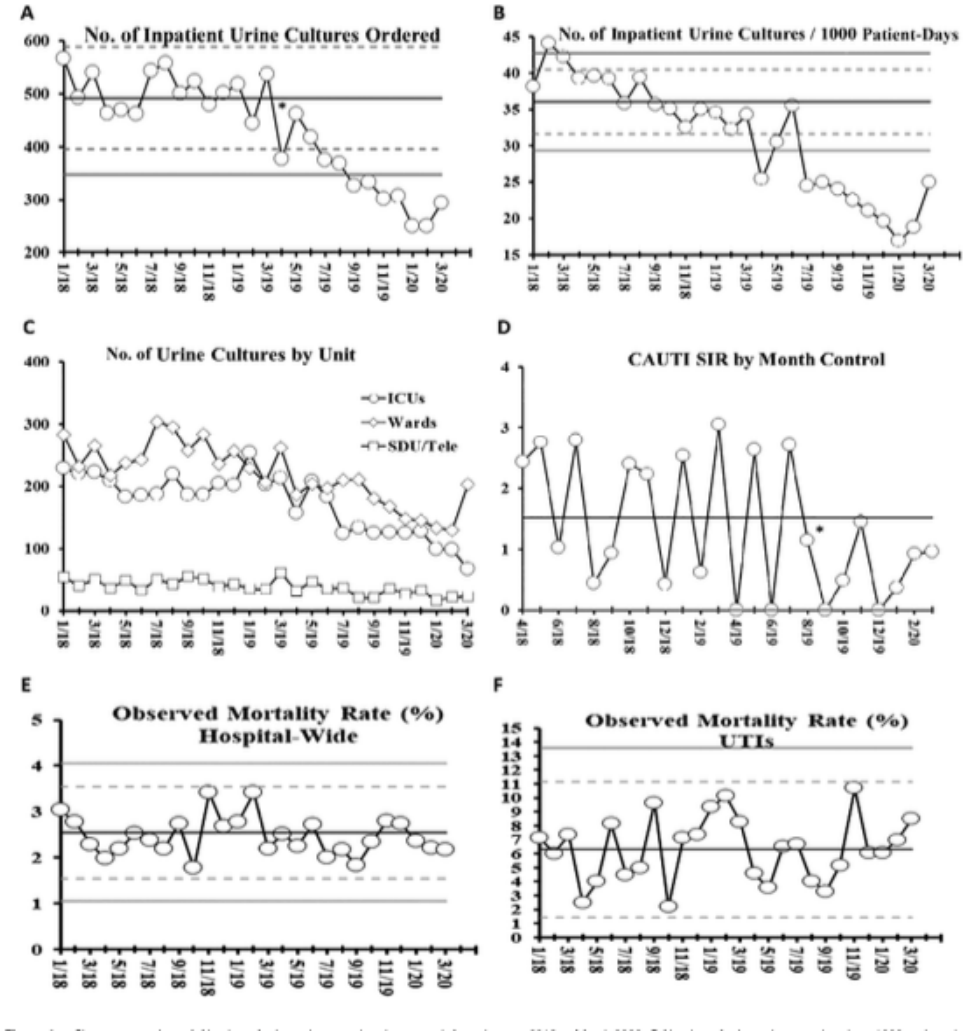


Figure 12: Results of a diagnostic stewardship intervention: reduced urine cultures, decreased catheter-associated UTIs over time, and no effect on mortality (Luu et al., 2021).

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5 Principle 2: Only Use Antibiotics When They Alter the Clinical Course

5.1 Antibiotics Are Not the Only Answer

The administration of antibiotics should not be a reflexive response to infection. It should be incorporated into an **overall, rational therapeutic plan**. Key considerations:

- Patients without bacterial infections **cannot** have their clinical course improved by antibiotics
- Some infections **require non-antibiotic interventions first** before antibiotics will be effective

5.1.1 Classic Example: Osteomyelitis with Exposed Bone

Figure 13: Osteomyelitis with persistently exposed bone — antibiotic therapy alone cannot resolve this infection. Surgical closure is required first; otherwise prolonged antibiotics will only select for multi-drug resistant organisms.

If the bone remains exposed, starting antibiotics will:

1. Fail to clear the infection
2. Select for increasingly resistant organisms
3. Leave the patient with an untreatable MDR bone infection

The proper approach: achieve **surgical closure/source control** first, then initiate appropriate antibiotic therapy.

5.2 Ethical Dilemmas

Two common ethical scenarios:

End-of-life care	Non-adherent HIV therapy
Should infections be treated aggressively in terminally ill patients? Generally yes for comfort, but the extent of treatment for MDR infections in terminal stages remains debated.	If a patient is non-adherent, antibiotic/antiviral therapy risks transforming their infection into a resistant, untreatable, and potentially transmissible disease.

6 Principle 3: Empirically Target Microbes in the Differential Diagnosis

6.1 Know the Spectrum of Activity

To select appropriate empiric therapy, clinicians must know which antibiotics cover which pathogens. The **Sanford Guide** (web.sanfordguide.com) provides comprehensive antimicrobial spectra tables using a grading system:

- ++ (double plus) — good activity with clinical evidence from trials
- + (green) — good *in vitro* activity expected to be effective, but possibly lacking clinical trial data for the specific indication
- ± (yellow/borderline) — resistance or pharmacokinetic concerns; unreliable
- **0** (red) — ineffective; do not use

6.1.1 The Impact of Resistance on Treatment Options

Consider the dramatic narrowing of treatment options with increasing resistance in *E. coli*:

Resistance phenotype	Available options	Key drugs remaining
Susceptible *E. coli*	Many options across multiple classes	Amoxicillin-clavulanate, piperacillin-tazobactam, cephalosporins, carbapenems, fluoroquinolones
ESBL-producing *E. coli* (25–30% of hospital isolates)	Dramatically reduced	Primarily **carbapenems**
KPC/MBL carbapenemase-producing *E. coli*	Only 1–2 options, none with strong clinical evidence	Newer agents (ceftazidime-avibactam, cefiderocol) with limited clinical data

6.2 Community-Acquired vs. Nosocomial Infections

The single most important question in selecting empiric therapy is: **where was this infection acquired?**

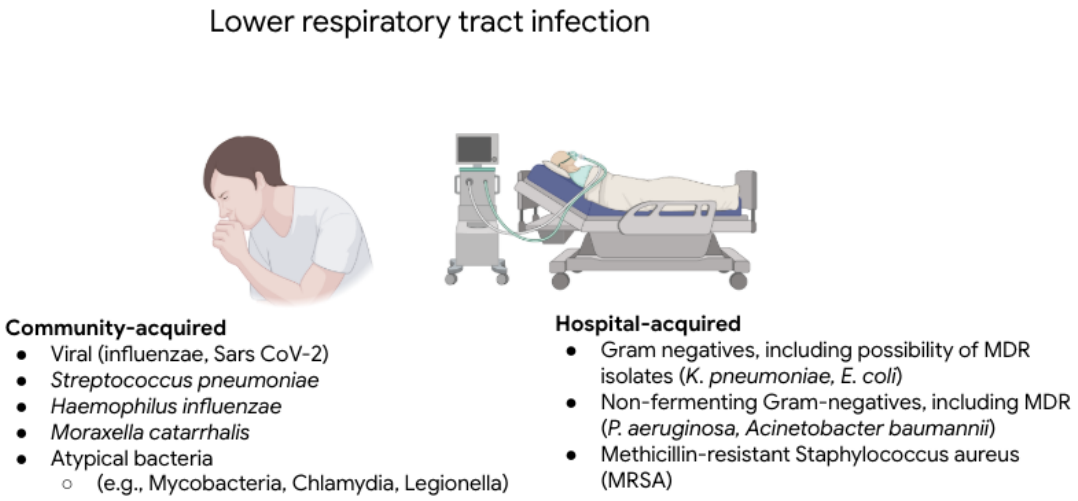

Figure 14: Pathogen differences between community-acquired and nosocomial respiratory infections.

Table 8: Key differences in empiric therapy based on site of infection acquisition.

Feature	Community infections	Nosocomial infections
MDR Gram-negatives	Infrequent	Common
MRSA risk	Low (unless specific risk factors)	High, especially with recent antibiotic use
Typical pathogens	*S. pneumoniae*, *H. influenzae*, *M. catarrhalis*, atypical pathogens	*Pseudomonas*, MRSA, non-fermenting Gram-negatives, Enterobacterales
Empiric approach	Narrow-spectrum oral therapy often sufficient	IV broad-spectrum therapy, consider ESBL/CRE coverage
Last-line agents	Reserve (e.g., fluoroquinolones)	May need as first-line

Exceptions

Community-acquired infections may be caused by *Pseudomonas* in patients with cystic fibrosis, bronchiectasis, chronic dialysis, indwelling catheters, or recent surgery.

---

7 Principle 4: Lower Threshold for Empirical Therapy in Critically Ill Patients

7.1 Antibiotic Timing Is Critical in Septic Shock

In critically ill patients with sepsis, there is no time for prolonged diagnostic deliberation. Observational evidence demonstrates that **each hour of delay** in administering appropriate antimicrobial therapy **increases the risk of death**.

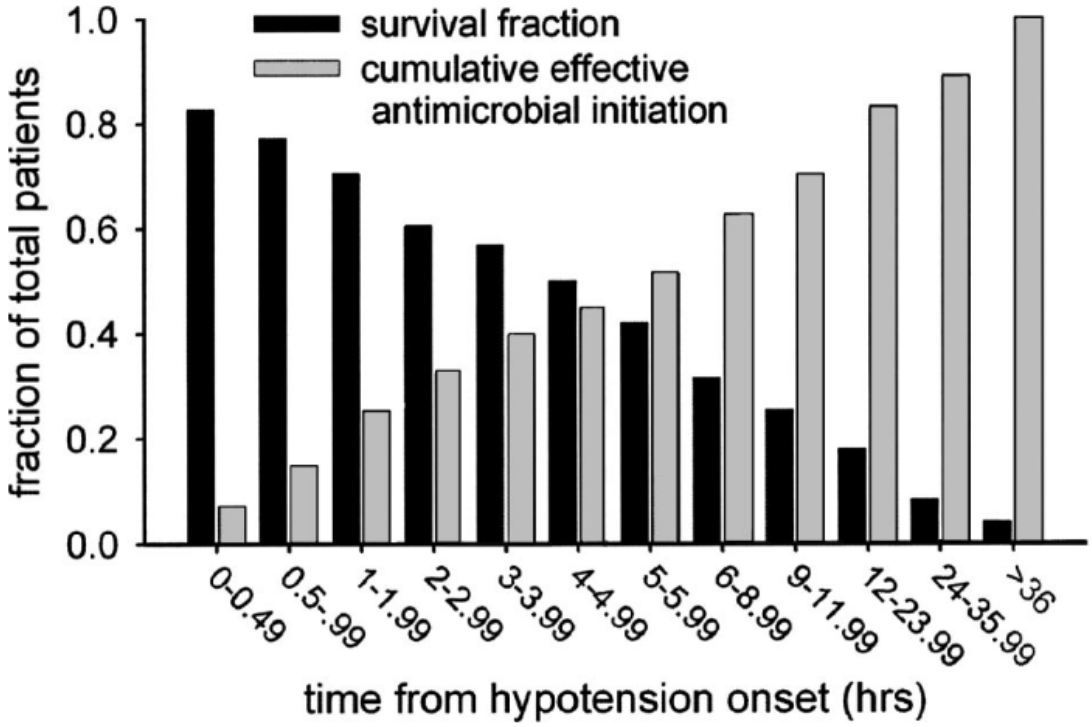

Figure 15: The Kumar study: survival fraction decreases as the time from hypotension onset to effective antibiotic administration increases. This landmark paper established the principle that antibiotic timing is critical in septic shock (Kumar et al., 2006).

7.1.1 Key Clinical Actions in Septic Shock

1. **Obtain blood cultures** before starting antibiotics — this is your best chance for a microbiological diagnosis (sensitivity drops after antibiotics are started)
2. **Start broad-spectrum empirical therapy immediately** — a carbapenem or other broad-spectrum regimen
3. **Ensure subsequent doses are administered on time** — the first AND second doses are critical
4. De-escalate once culture results are available

> ⚠ Clinical Pearl
>
> In septic patients, you may need to start broader empirical therapy than you would in a stable patient. The risk of undertreating a critically ill patient with an inadequate antibiotic far outweighs the collateral damage of temporary broad-spectrum therapy.

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8 Principle 5: Host Factors Affect the Spectrum of Empirical Therapy

8.1 Immunocompromise Changes the Differential Diagnosis

The "flavor" of immunosuppression determines the types of pathogens that must be considered:

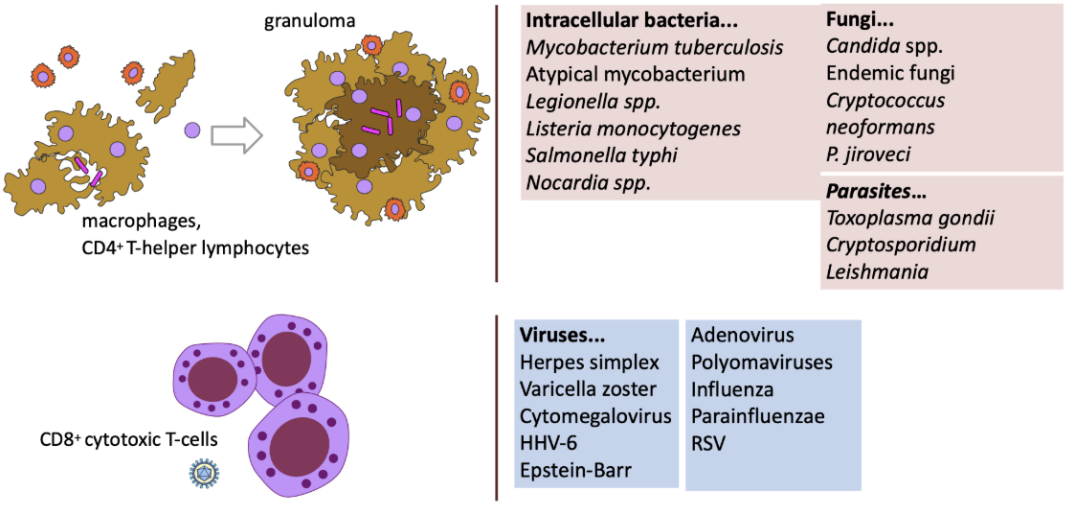


Figure 16: Impact of immunocompromise on the spectrum of potential pathogens — cell-mediated immunity defects broaden the differential to include intracellular bacteria, fungi, and parasites.

Key question: Does this patient have impaired **cell-mediated immunity**? If so:

- **Acute presentation** → viral infections, common bacteria, **intracellular bacteria** (Legionella, Listeria, Salmonella Typhi, Nocardia)
- **Delayed/subacute presentation** → **fungal infections** (Candida, Aspergillus), parasites
- Standard -lactams **will not cover** intracellular pathogens

8.1.1 Common Immunocompromising Conditions

1. Chronic diseases (type 1 diabetes, COPD)
2. Autoimmune diseases (lupus, rheumatoid arthritis)
3. Primary immunodeficiencies
4. Cancer and/or chemotherapy
5. HIV
6. Solid organ or bone marrow transplant
7. Advanced age
8. Malnutrition
9. **Chronic corticosteroid use or other immunosuppressive medications**
10. Chronic infections
11. Smoking (impaired ciliary response)

8.2 Glucocorticoids: The "Credit Cards" of Immunosuppressive Therapy

Corticosteroids provide **immediate anti-inflammatory benefit** (like a credit card) but accrue a compounding "debt" of immunosuppression — the bill comes due as **opportunistic infections**.

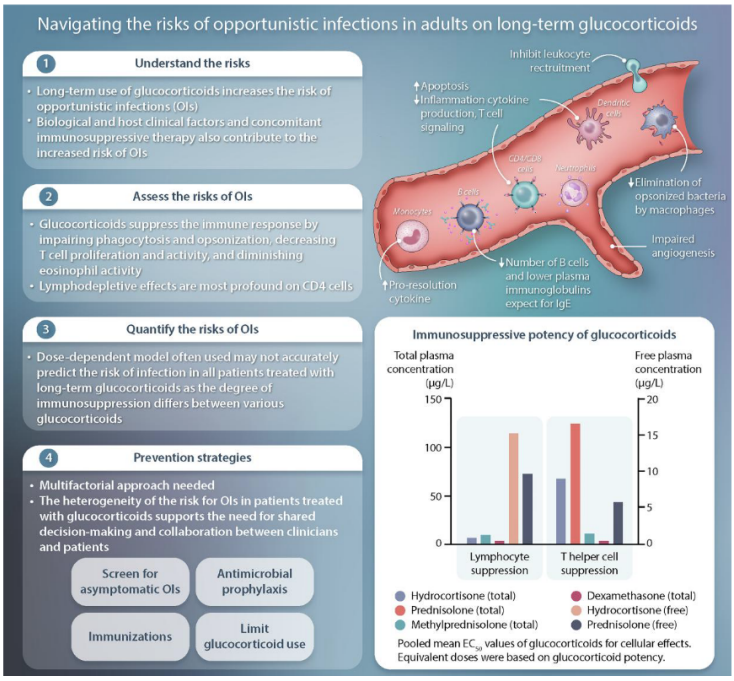
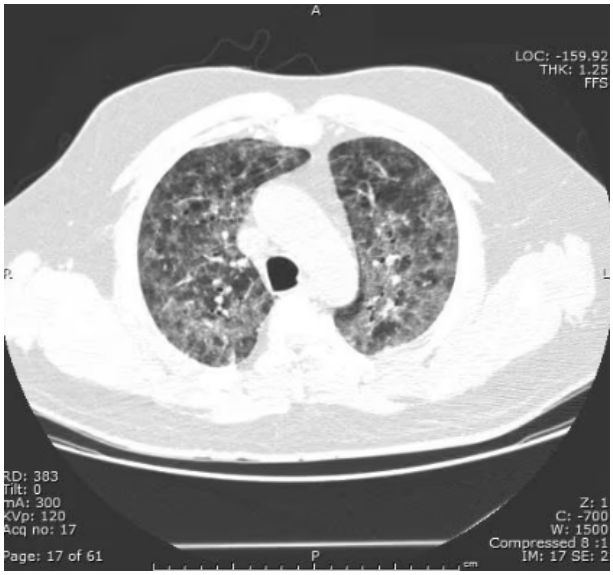


Figure 17: Dose-dependent risk of opportunistic infections with glucocorticoid use — significant risk begins at >10 mg prednisone equivalents (PEQ) per day for 2–4 weeks (Chastain et al., 2023).

8.2.1 Case Example: Hidden Immunosuppression

A patient with **pulmonary sarcoidosis** on 10 mg/day prednisone presents with community-acquired pneumonia (diffuse bilateral infiltrates). Initial empiric therapy for typical CAP (*S. pneumoniae*) is started, but the patient deteriorates. Bronchoscopy eventually reveals ***Pneumocystis jirovecii* pneumonia (PCP)** — an opportunistic infection typically seen in bone marrow transplant recipients or advanced HIV patients, but here caused by chronic low-dose steroid use.

(a) Chest X-ray: diffuse bilateral infiltrates compatible with either typical CAP or PCP.



(a) CT scan: patchy areas of ground-glass attenuation — characteristic of PCP.

Clinical Pearl

Always ask about corticosteroid use — even low doses (>10 mg prednisone equivalents daily) for prolonged periods can cause clinically significant immunosuppression. Consider PCP prophylaxis in patients on prolonged steroids. For a comprehensive review, see Chastain et al., *Clin Infect Dis* 2023 (Chastain et al., 2023).

---

9 Principle 6: Use PK/PD Principles to Optimize Treatment Selection and Dosing

9.1 It's Not Enough to Pick the Right Drug

Selecting an antibiotic with appropriate **pharmacodynamic** (PD) activity — the right spectrum — is only half the equation. You must also consider the **pharmacokinetic** (PK) properties: does the drug reach the site of infection at adequate concentrations?

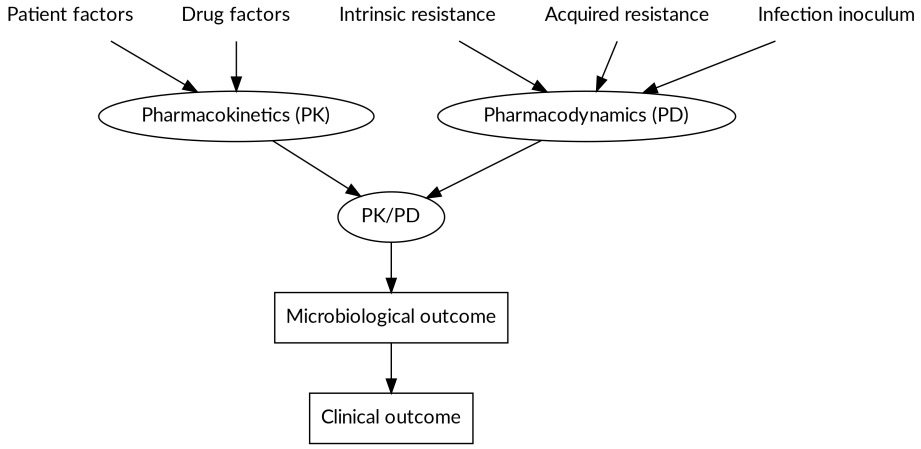


Figure 20: The PK/PD framework: patient and drug factors determine pharmacokinetics; intrinsic/acquired resistance and inoculum determine pharmacodynamics. Their interaction (PK/PD) predicts microbiological and clinical outcomes.

Pharmacodynamics (spectrum, MIC, potency) — you **cannot control** whether the pathogen is intrinsically or acquired resistant.

Pharmacokinetics (absorption, distribution, metabolism, elimination) — you **can control** the dose, route, and administration schedule to maximize drug exposure at the infection site.

9.2 Key Pharmacokinetic Variables

9.2.1 The Concentration-Time Curve

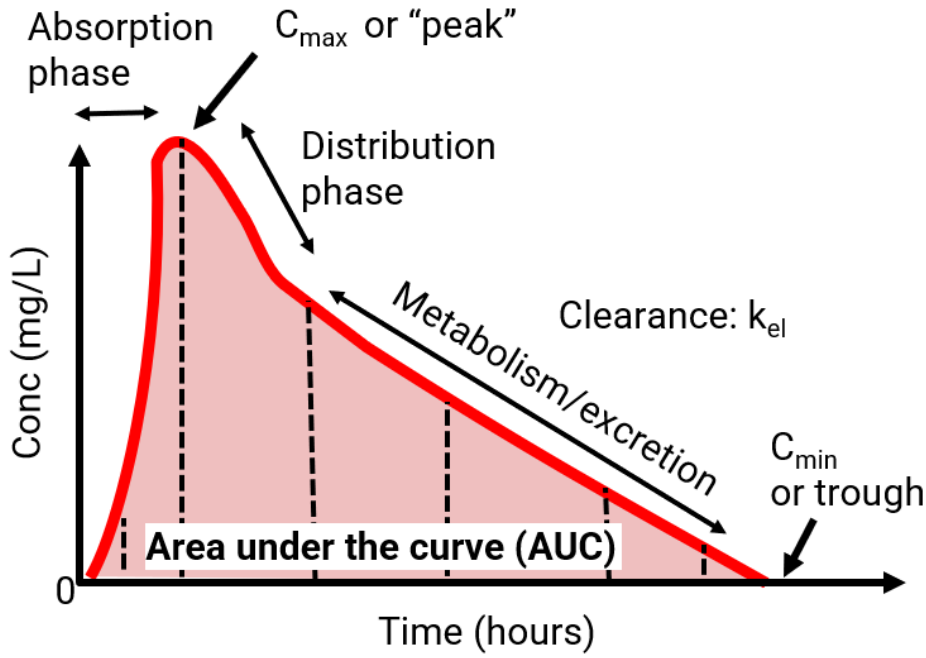


Figure 21: Key pharmacokinetic parameters: C_{max} (peak concentration), C_{min} (trough concentration), AUC (area under the curve — a measure of total drug exposure), and the relationship of drug concentrations to the MIC of the pathogen.

9.2.2 Volume of Distribution (Vd)

The Vd is a **theoretical volume** that relates the total amount of drug in the body to the plasma concentration. It is estimated, not directly measured, and reported in liters (L) or liters per kilogram (L/kg).

- **Average plasma volume** in adults 3 L
- **Plasma + interstitial fluid** 13–14 L
- **Total body water** 20 L (plasma + extracellular water)

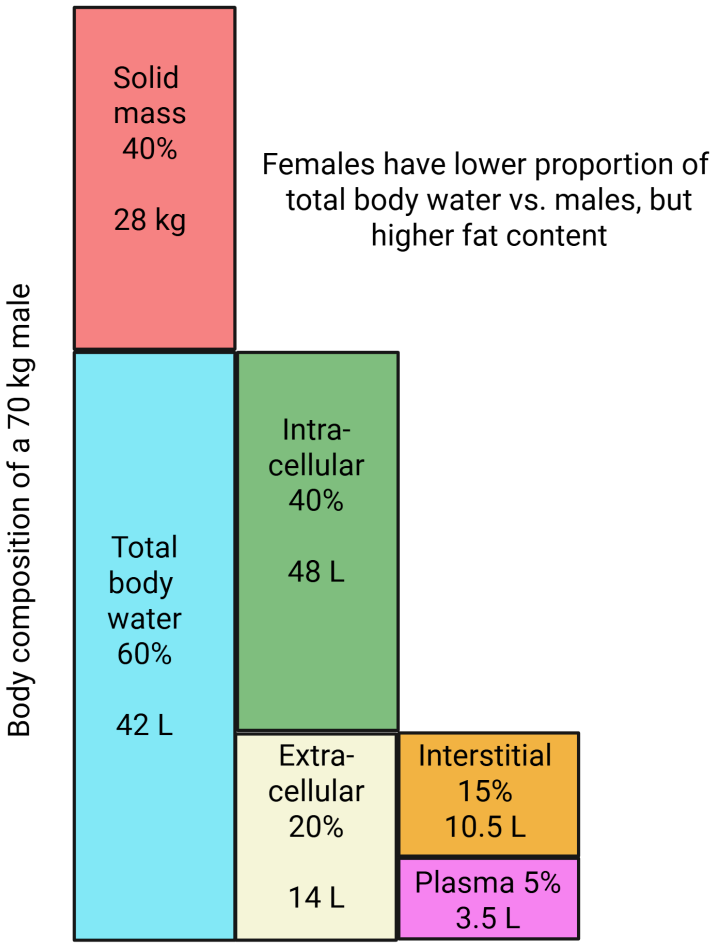


Figure 22: Breakdown of body fluid compartments and their relationship to volume of distribution.

9.2.2.1 Clinical Significance of Vd

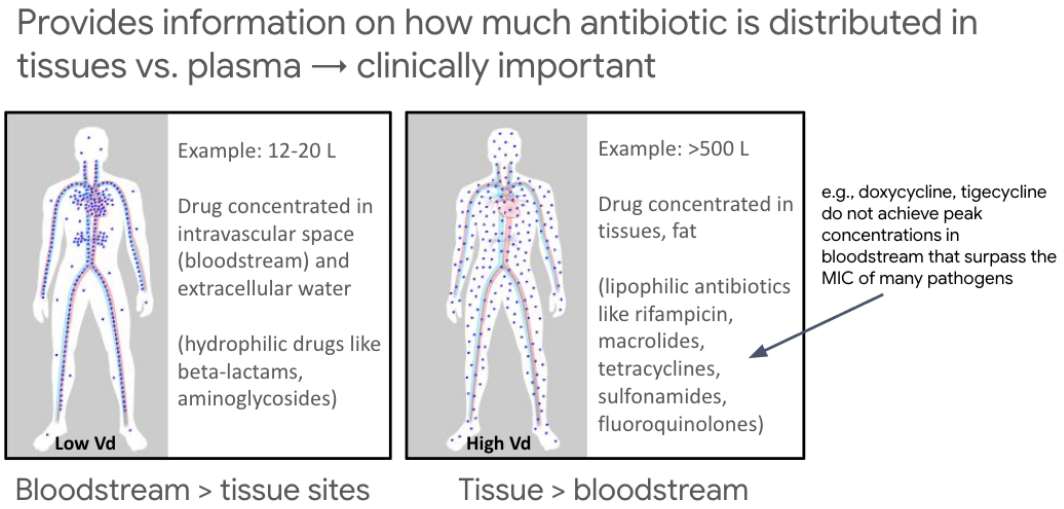
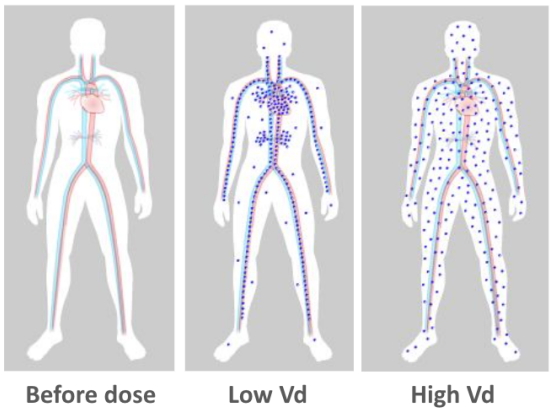


Figure 24: Clinical relevance of Vd for antibiotic selection: drugs with high Vd (macrolides, doxycycline, tigecycline, rifampin) should NOT be used as monotherapy for bloodstream infections, even if susceptibility testing says "susceptible," due to low blood concentrations and higher failure/resistance rates.



(a) Relationship between Vd and serum vs. tissue drug concentrations.

- **Low Vd** (20 L): Drug concentrates in the bloodstream → **better for bloodstream infections** (e.g., -lactams, aminoglycosides)
- **High Vd** (>46 L): Drug is sequestered in tissues (fat, skin, nails) → serum concentrations will be **low** → poor choice for bloodstream infections, but may be useful for tissue infections

Interpretation:

> **!** Key Rule of Thumb
>
> - **Low Vd drugs** (-lactams, aminoglycosides, vancomycin) → preferred for **bloodstream infections, endocarditis**
> - **High Vd drugs** (macrolides, doxycycline, tigecycline, rifampin) → useful for **pneumonia, deep tissue infections** but NOT for high-grade bacteremia
>
> Historical failures: early endocarditis studies showed poor outcomes with tetracyclines and macrolides (high Vd, low blood levels) compared to penicillins (low Vd, high blood levels) — this led to the erroneous conclusion that "bactericidal" drugs are inherently superior to "bacteriostatic" drugs, when the real issue was pharmacokinetics.

9.2.2.2 Vd Changes in Disease States

Vd is not static — it can be altered by:

- Body mass, age, sex, pregnancy
- Kidney disease, liver disease
- Drug interactions (displacement of protein binding)
- **Critical illness and sepsis**

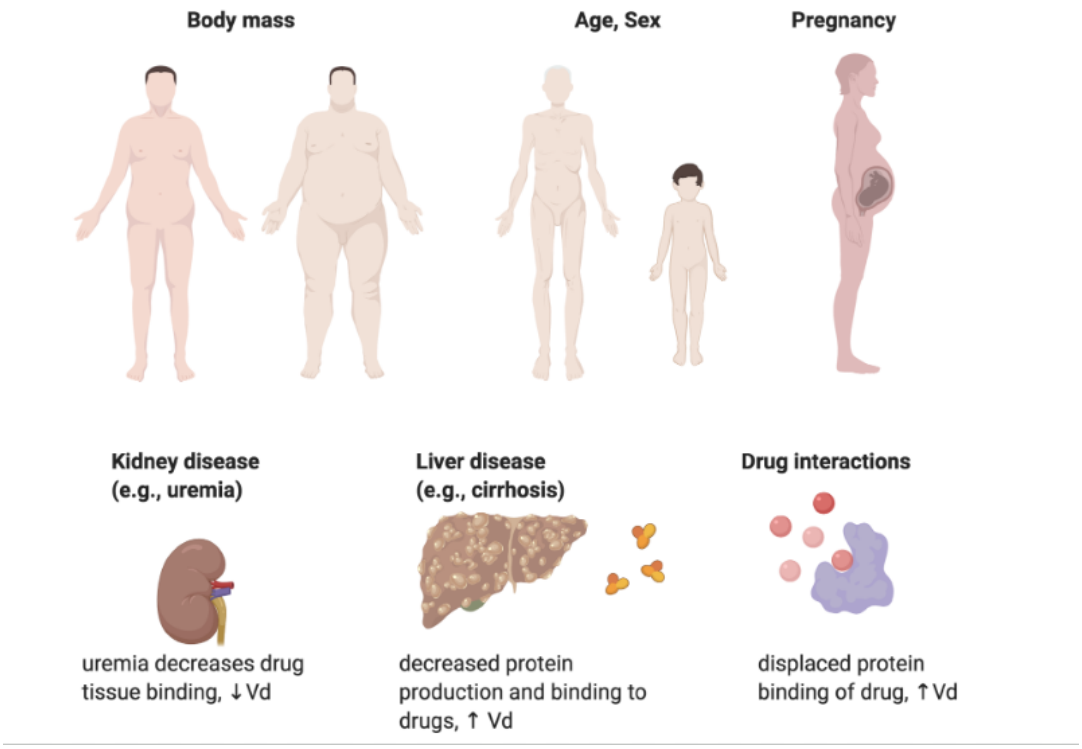


Figure 25: Factors that alter Vd in various clinical conditions.

9.2.2.3 Vd Changes in Critical Illness — A Major Clinical Problem

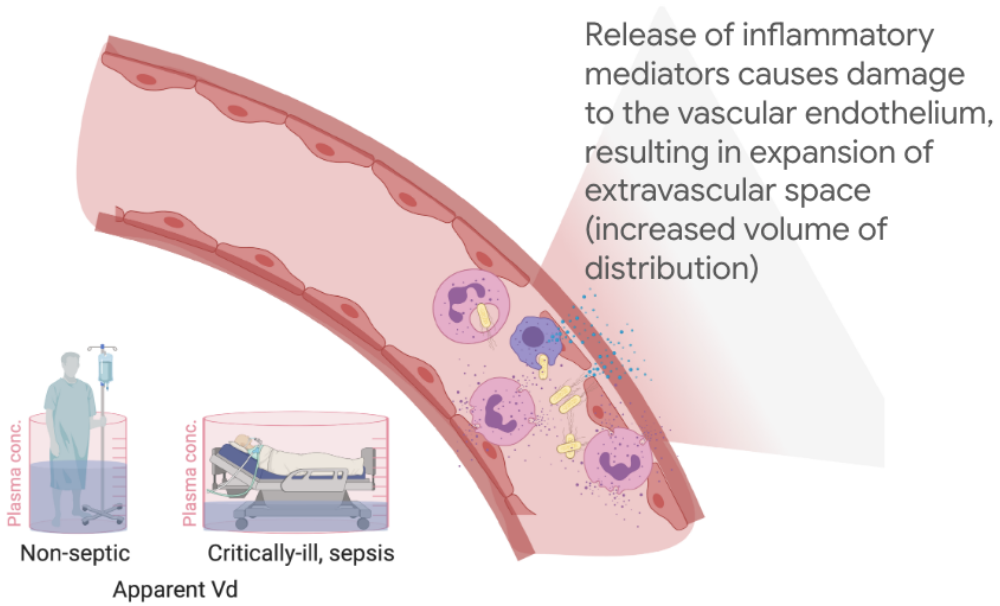


Figure 26: In sepsis, disseminated inflammation, fibrin deposition, and fluid extravasation dramatically increase Vd, resulting in lower-than-expected blood concentrations of antibiotics.

Practical consequence: In septic patients, the "beaker" (Vd) gets much bigger. A standard dose that would produce adequate concentrations in a non-septic patient may result in **subtherapeutic levels**, contributing to treatment failure and resistance emergence.

9.2.3 The DALI Study: ICU Underdosing Is the Rule, Not the Exception

A landmark multicenter study across 60–65 ICUs in Europe and Australia sampled -lactam concentrations in critically ill patients at leading academic medical centers:

- **>1/3 of patients** did not achieve even the **minimum** PK/PD target for efficacy
- **>2/3 of patients** did not achieve the more aggressive target needed to prevent resistance emergence

> ⚠ Clinical Implication
>
> We routinely underdose critically ill patients with sepsis — the exact population where our biggest resistance problems occur. This is because antibiotic doses were established in studies of patients with urinary tract infections or mild intra-abdominal infections, not in septic ICU patients.

9.3 Drug Penetration and Site-Specific Considerations

(Roberts et 2014)

9.3.1 Ventilator-Associated Pneumonia

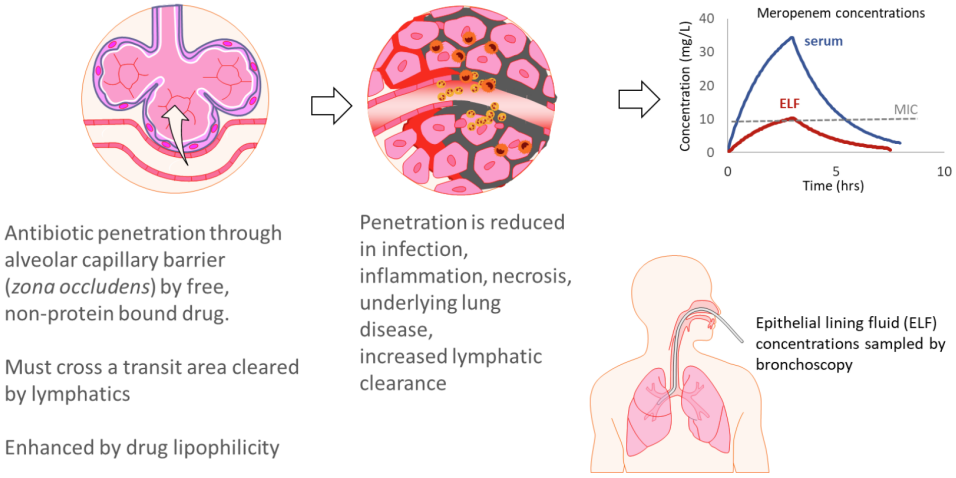


Figure 27: Meropenem concentrations in serum versus epithelial lining fluid (ELF) — lung concentrations barely surpass pathogen MICs, contributing to resistance emergence in VAP.

The lungs represent one of the most challenging sites for antibiotic penetration, which is why **ventilator-associated pneumonia** is one of the most common settings for resistance emergence.

9.3.2 Anatomically Privileged Sites

Certain body compartments have barriers that restrict antibiotic penetration:

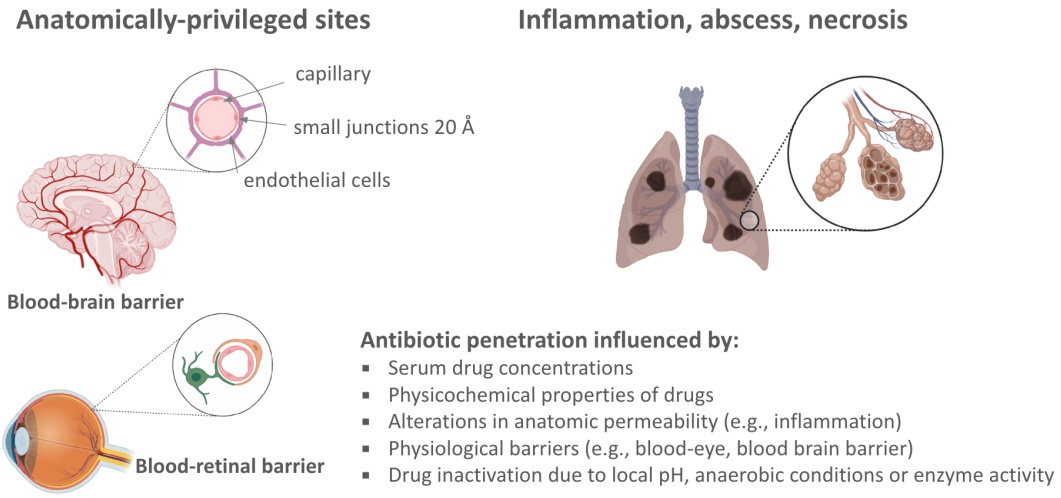


Figure 28: Anatomically privileged sites where antibiotics may not achieve therapeutic concentrations — CNS (meninges), eye (vitreous), prostate, bone — requiring higher doses or specific drug selection.

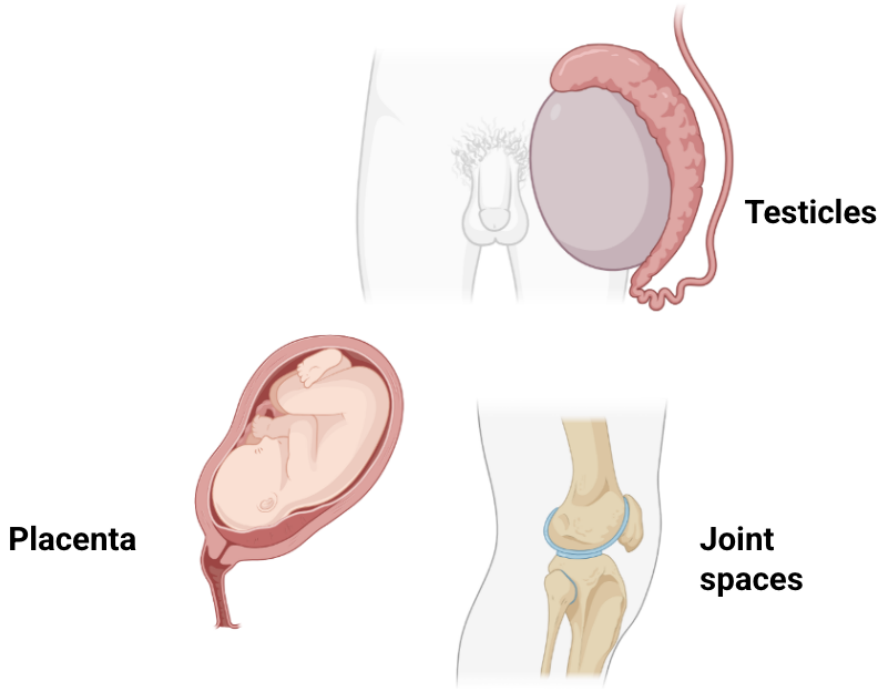


Figure 29: Additional examples of restricted penetration sites and the impact of necrosis and abscess cavities on antibiotic efficacy.

9.3.3 Using Site-Specific Pharmacokinetics to Advantage

Not all pharmacokinetic barriers work against us. Some sites **concentrate** antibiotics:

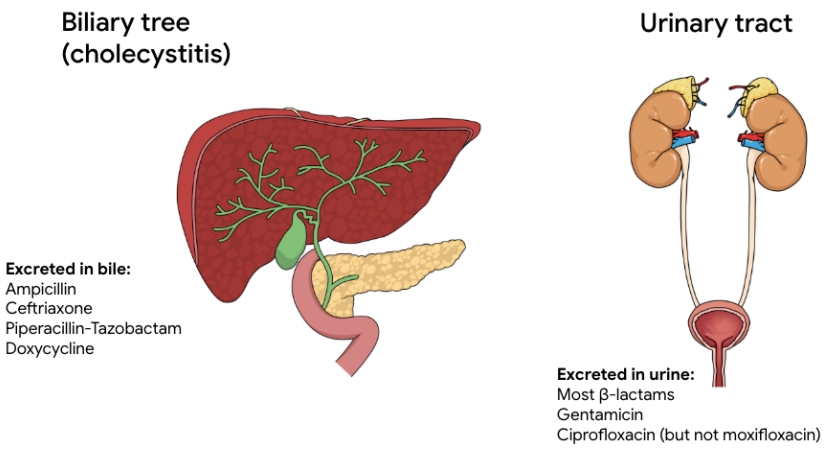


Figure 30: Match the antibiotic to the site of infection — some drugs concentrate in bile (useful for cholecystitis) or urine (useful for UTI).

9.3.3.1 Urinary Tract: A Major Advantage

Many antibiotics are excreted through the kidneys and achieve **urinary concentrations vastly exceeding serum levels**. This means some antibiotics may be effective for UTIs even when the laboratory reports "resistant," because the urinary concentrations surpass the MIC.

Table 1. Serum and urine concentrations of selected antimicrobials[a]

Drug	Dose, Route	Excretion *via*	Peak Serum Concentration (μg/ml)	% Renal Excretion of Unchanged Drug	Urine Concentration (μg/ml)
β-Lactams					
ampicillin	0.5 g p.o.	GF, TS	3 to 6	75 to 92	1000 to 2250
amoxicillin	0.5 g p.o.	GF, TS	6 to 10	60 to 98	300 to 1300
aztreonam	1 g IV	GF, TS	125	65 to 94	1000 to 5000
piperacillin	3 g IV	GF, TS	209	74 to 89	8000
ticarcillin	3 g IV	GF, TS	257	80 to 99	650 to 2500
cephalexin	1 g p.o.	GF, TS	32	91 to 100	5000 to 10,000
cefuroxime	1 g p.o.	GF, TS	14	50	1000 to 7000
cefixime	0.2 g p.o.		1 to 4	20 to 50	
cefdinir	0.3 g p.o.		1.6	18	21 to 139
cefpodoxime	0.2 g p.o.		2.3	29 to 33	20
cefazolin	1 g IV	GF, TS	188	90 to 96	700 to 2000
ceftriaxone	1 g IV	GF	130	65 to 95	549 to 995
imipenem	0.5 g IV	GF, TS	21 to 58	5 to 42	500
meropenem	1 g IV	GF, TS	50	62 to 83	N/D
ertapenem	1 g IV		154	38	N/D
Fluoroquinolones					
ciprofloxacin	0.5 g p.o.	GF, TS	1.6 to 2.9	30 to 50	350
gatifloxacin	0.4 g p.o.		4.2	65	N/D
gemifloxacin	0.320 g p.o.		0.7 to 2.6	<35	N/D
levofloxacin	0.5 g p.o.	GF	5.7	61 to 86	286
moxifloxacin	0.4 g p.o.		4.5	20	N/D
Miscellaneous					
nitrofurantoin	0.1 gm p.o.	GF, TS, TR	<2	27 to 56	50 to 200
trimethoprim sulfamethoxazole	0.16/0.8 g p.o.		9/105	50 to 75/10 to 30	31 to 165/10 to 133
trimethoprim	0.2 gm p.o.	GF, TS	2	50 to 75	70 to 100
sulfamethoxazole	0.8 g p.o.	GF, TS, TR	46	N/D	400 to 2000
Aminoglycosides					
amikacin	0.5 g IV	GF, TR	17 to 25	>90	170 to 1720
gentamicin	0.08 g IV	GF, TR	4 to 8	>90	400 to 500
tobramycin	0.08 g IV	GF, TR	4 to 8	>90	94 to 443

[a]The urine concentration is expressed as the range of concentrations achieved after a standard dose in patients with normal renal function. Single values represent the average urine concentration over a standard dosage interval (21–23). GF, glomerular filtration; IV, intravenously; N/D, no data; p.o., orally; TR, tubular reabsorption; TS, tubular secretion.

Figure 31: Comparison of serum vs. urinary concentrations for common antibiotic classes — urinary concentrations are often 100–1000× higher than serum (Gilbert, 2006).

Table 9: Antibiotic concentrations achievable in the urine far exceed serum levels, allowing treatment of resistant UTIs with drugs that might appear "resistant" based on serum-derived breakpoints.

Antibiotic class	Serum concentration	Urinary concentration
Fluoroquinolones	1.6–2.9 µg/mL	~350 µg/mL
-lactams	3–6 µg/mL	1,000–2,250 µg/mL
Aminoglycosides	4–8 (or 17–25) µg/mL	Hundreds to thousands µg/mL

9.4 Clearance: The Second Key PK Variable

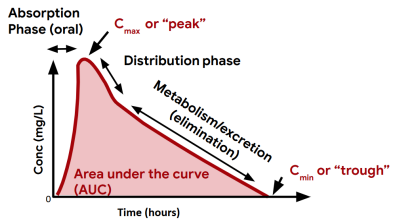


Figure 32: Clearance — drug elimination from the body, described as volume of blood cleared of drug per unit time (mL/min or L/hr). Total body clearance = $CL_{renal} + CL_{hepatic} + CL_{other}$.

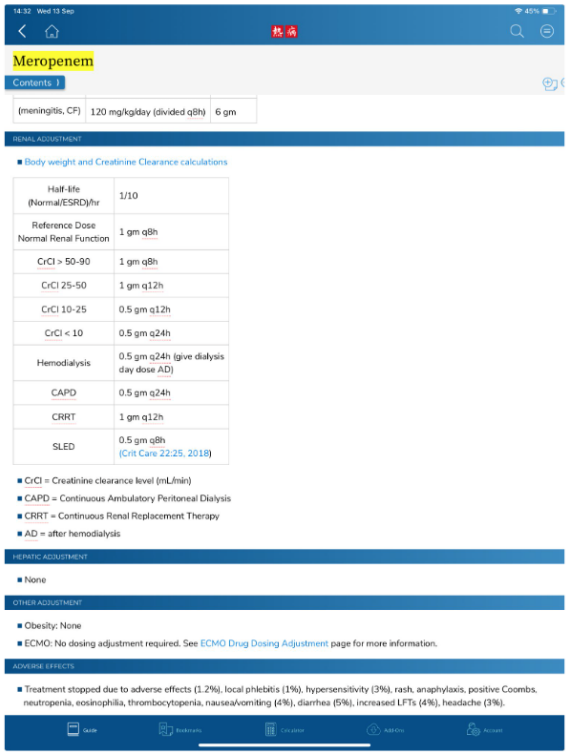
9.4.1 Vd vs. Clearance: Critical Distinctions

Parameter	Determines	Clinical use
Volume of distribution (Vd)	How much drug distributes into tissues	**Loading dose** — fill the "tank"
Clearance (CL)	How fast the body eliminates drug	**Maintenance dose** — how often and how much to give

- Vd and CL are both physiologically based but **do not directly interact** with each other
- A change in Vd does not change clearance, and vice versa
- **CL is NOT used to determine the initial (loading) dose** — only Vd matters for that

9.4.2 Renal Dose Adjustment

Most antibiotics are eliminated via the kidneys, and maintenance doses must be adjusted for renal function.



Meropenem

Contents

(meningitis, CF)	120 mg/kg/day (divided q8h)	6 gm

RENAL ADJUSTMENT

- Body weight and Creatinine Clearance calculations

Half-life (Normal/ESRD)/hr	1/10
Reference Dose Normal Renal Function	1 gm q8h
CrCl > 50-90	1 gm q8h
CrCl 25-50	1 gm q12h
CrCl 10-25	0.5 gm q12h
CrCl < 10	0.5 gm q24h
Hemodialysis	0.5 gm q24h (give dialysis day dose AD)
CAPD	0.5 gm q24h
CRRT	1 gm q12h
SLED	0.5 gm q8h (Crit Care 22:25, 2018)

- CrCl = Creatinine clearance level (mL/min)
- CAPD = Continuous Ambulatory Peritoneal Dialysis
- CRRT = Continuous Renal Replacement Therapy
- AD = after hemodialysis

HEPATIC ADJUSTMENT

- None

OTHER ADJUSTMENT

- Obesity: None
- ECMO: No dosing adjustment required. See ECMO Drug Dosing Adjustment page for more information.

ADVERSE EFFECTS

- Treatment stopped due to adverse effects (1.2%), local phlebitis (1%), hypersensitivity (3%), rash, anaphylaxis, positive Coombs, neutropenia, eosinophilia, thrombocytopenia, nausea/vomiting (4%), diarrhea (5%), increased LFTs (4%), headache (3%).

Figure 33: Example: meropenem dose adjustment based on estimated creatinine clearance.

9.4.3 Estimating Renal Function

The **Cockcroft-Gault** formula is commonly used:

$$\text{CrCl (mL/min)} = \frac{(140 - \text{age}) \times \text{weight (kg)}}{72 \times \text{SCr (mg/dL)}} \times 0.85 \text{ (if female)}$$

Important limitations:

- Developed primarily in **Caucasian males** with **chronic** renal disease
- Does not account for older age, comorbidities, or drug interactions with renal tubular secretion
- Not validated in patients with **acute kidney injury** or sepsis

9.4.4 Critical Problem: Premature Renal Dose Reduction

The Two Most Common Dosing Mistakes

1. **Reducing the FIRST dose** based on renal function — the first dose should be based on **Vd** (to fill the "tank"), NOT on clearance. Never reduce the loading dose.
2. **Reducing doses in patients with REVERSIBLE renal dysfunction** — up to 50% of patients with acute infection have **acute** (not chronic) renal impairment that frequently resolves within 48 hours with fluid resuscitation. Premature dose reduction leads to subtherapeutic antibiotic concentrations during the most critical treatment window.

Antibiotic renal dose adjustments in drug labels are based on patients with **chronic kidney disease**. However, in infectious disease:

- Renal dysfunction is frequently **acute**, not chronic
- Serum creatinine changes are **delayed** — creatinine decreases lag behind actual improvement in kidney function
- Using serum creatinine to dose antibiotics in acute illness is like **"driving by only looking in the rearview mirror"** (Crass et al., 2019)

Practical guidance: If a patient presents severely dehydrated and hypotensive with an elevated creatinine, and you believe this is acute and reversible with fluids, **do not reduce antibiotic doses in the first 48 hours** (especially for safe antibiotics like -lactams). The risk of undertreating the infection during the critical early period outweighs the risk of a brief period of higher drug levels.

Loading Dose Principle

The **first dose** of an antibiotic in a septic patient is determined by **Vd** (to fill the expanded volume). **Clearance** determines subsequent maintenance doses. Even in patients with renal dysfunction — give the **full first dose**, then adjust maintenance based on kidney function trends.

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10 Summary of Key Concepts (Slides 1–62)

This section of the lecture establishes six foundational principles for antibiotic therapy:

1. **The medical history drives everything** — from diagnostic test selection to empiric therapy choices. The brucellosis case illustrates how history completely redirects the approach.
2. **Antibiotics are powerful but not benign** — 1 in 5 patients experience adverse events; they should only be used when they can alter clinical outcomes. Source control must precede or accompany antibiotic therapy.
3. **Know your bugs and drugs** — spectrum of activity varies dramatically based on resistance phenotype. Community vs. nosocomial acquisition fundamentally changes pathogen probability.
4. **Time kills in sepsis** — every hour of delayed appropriate therapy increases mortality. Start broad, obtain cultures first, then de-escalate.
5. **Immunosuppression broadens the differential** — particularly cell-mediated immunity defects. Corticosteroids are a "hidden" source of immunosuppression that can lead to unexpected opportunistic infections.
6. **Pharmacokinetics matter as much as spectrum** — Vd determines where the drug goes (loading dose), clearance determines how long it stays (maintenance dose). ICU underdosing is endemic and contributes to resistance. Match the drug's pharmacokinetic profile to the site of infection.

---

11 References

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